

# Chapter 4 Exam Review

Name \_\_\_\_\_

**SHORT ANSWER.** Write the word or phrase that best completes each statement or answers the question.

Approximate the number using a calculator. Round your answer to three decimal places.

1)  $5^{1.2}$  1) \_\_\_\_\_

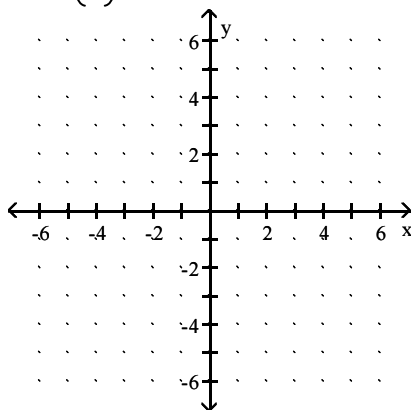
2)  $3\sqrt{5}$  2) \_\_\_\_\_

**Solve the problem.**

- 3) A city is growing at the rate of 0.6% annually. If there were 3,812,000 residents in the city in 1994, find how many (to the nearest ten-thousand) are living in that city in 2000. Use  $y = 3,812,000(2.7)^{0.006t}$ . 3) \_\_\_\_\_

**Graph the function by making a table of coordinates.**

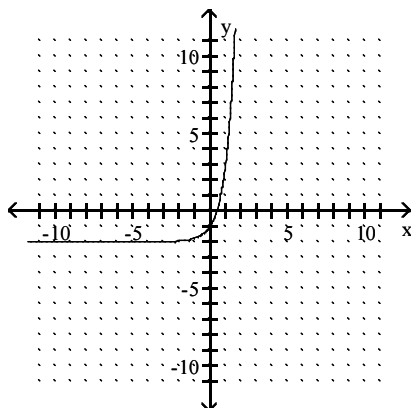
4)  $f(x) = \left(\frac{1}{2}\right)^x$  4) \_\_\_\_\_



**MULTIPLE CHOICE.** Choose the one alternative that best completes the statement or answers the question.

The graph of an exponential function is given. Select the function for the graph from the functions listed.

5) 5) \_\_\_\_\_



A)  $f(x) = 5^x$

B)  $f(x) = 5^x + 2$

C)  $f(x) = 5^x - 2$

D)  $f(x) = 5^x - 2$

**SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.**

Use the compound interest formulas  $A = P\left(1 + \frac{r}{n}\right)^{nt}$  and  $A = Pe^{rt}$  to solve.

- 6) Find the accumulated value of an investment of \$15,000 at 8% compounded quaterly for 12 years. 6) \_\_\_\_\_

**Write the equation in its equivalent exponential form.**

7)  $\log_5 x = 3$  7) \_\_\_\_\_

8)  $\log_b 256 = 4$  8) \_\_\_\_\_

**Write the equation in its equivalent logarithmic form.**

9)  $4^2 = 16$  9) \_\_\_\_\_

10)  $4^{-2} = \frac{1}{16}$  10) \_\_\_\_\_

11)  $\sqrt[3]{343} = 7$  11) \_\_\_\_\_

**Evaluate the expression without using a calculator.**

12)  $\log_4 64$  12) \_\_\_\_\_

13)  $\log_6 1$  13) \_\_\_\_\_

14)  $\log_9 9$  14) \_\_\_\_\_

15)  $\log_8 8^{19}$  15) \_\_\_\_\_

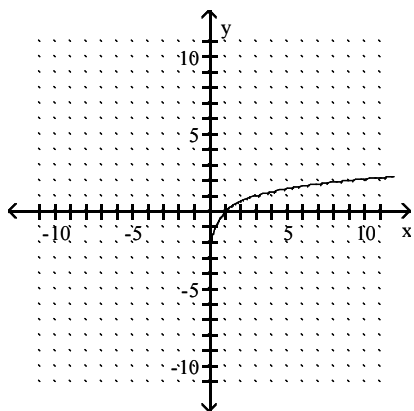
16)  $9^{\log_9 16}$  16) \_\_\_\_\_

**MULTIPLE CHOICE.** Choose the one alternative that best completes the statement or answers the question.

The graph of a logarithmic function is given. Select the function for the graph from the options.

17)

17) \_\_\_\_\_



A)  $f(x) = \log_3(x + 1)$

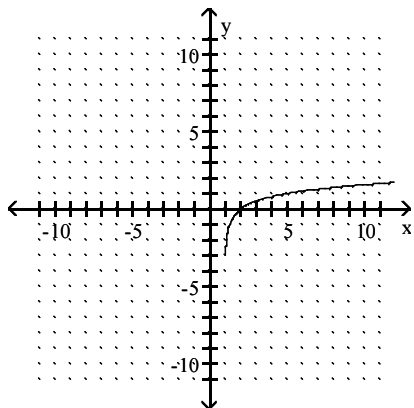
B)  $f(x) = \log_3 x + 1$

C)  $f(x) = \log_3 x$

D)  $f(x) = \log_3(x - 1)$

18)

18) \_\_\_\_\_



A)  $f(x) = \log_4 x - 1$

B)  $f(x) = \log_4(x - 1)$

C)  $f(x) = \log_4 x$

D)  $f(x) = \log_4(x + 1)$

**SHORT ANSWER.** Write the word or phrase that best completes each statement or answers the question.

Find the domain of the logarithmic function.

19)  $f(x) = \log_9(x + 4)$

19) \_\_\_\_\_

20)  $f(x) = \ln(9 - x)$

20) \_\_\_\_\_

Use properties of logarithms to expand the logarithmic expression as much as possible. Where possible, evaluate logarithmic expressions without using a calculator.

21)  $\log_6(7 \cdot 11)$

21) \_\_\_\_\_

22)  $\log_2(8x)$

22) \_\_\_\_\_

23)  $\log_5 \left( \frac{7}{13} \right)$

23) \_\_\_\_\_

24)  $\ln \left( \frac{e^6}{11} \right)$

24) \_\_\_\_\_

25)  $\log_2 7^{-2}$

25) \_\_\_\_\_

26)  $\ln \sqrt[4]{x}$

26) \_\_\_\_\_

Use properties of logarithms to condense the logarithmic expression. Write the expression as a single logarithm whose coefficient is 1. Where possible, evaluate logarithmic expressions.

27)  $2 \log_x 3 + \log_x 3$

27) \_\_\_\_\_

28)  $\log_5 500 - \log_5 4$

28) \_\_\_\_\_

29)  $\ln x + 7 \ln y$

29) \_\_\_\_\_

30)  $\frac{1}{2} \log_4 x + \log_4 y$

30) \_\_\_\_\_

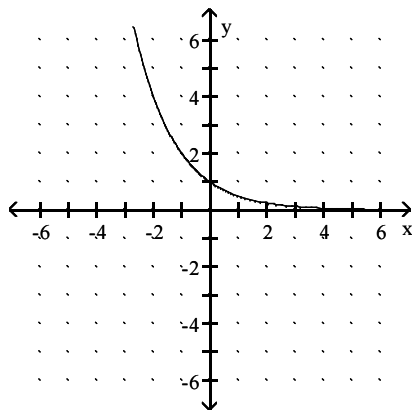
31)  $\ln x + 8 \ln y$

31) \_\_\_\_\_

# Answer Key

Testname: MTH\_131\_REVIEW\_CH4A\_EXAM

- 1) 6.899
- 2) 11.665
- 3) 3,950,000
- 4)



- 5) D
- 6) \$38,449.56
- 7)  $5^3 = x$
- 8)  $b^4 = 256$
- 9)  $\log_4 16 = 2$
- 10)  $\log_4 \frac{1}{16} = -2$
- 11)  $\log_{343} 7 = \frac{1}{3}$
- 12) 3
- 13) 0
- 14) 1
- 15) 19
- 16) 16
- 17) C
- 18) B
- 19)  $(-4, \infty)$
- 20)  $(-\infty, 9)$
- 21)  $\log_6 7 + \log_6 11$
- 22)  $3 + \log_2 x$
- 23)  $\log_5 7 - \log_5 13$
- 24)  $6 - \ln 11$
- 25)  $-2 \log_2 7$
- 26)  $\frac{1}{4} \ln x$
- 27)  $\log_x 27$
- 28) 3
- 29)  $\ln xy^7$
- 30)  $\log_4 y \sqrt{x}$
- 31)  $\ln xy^8$